Pure Mathematics

APPLIED TECHNOLOGY SCHOOLS

(First Secondary) – First Semester



Part One

Introduction

The Arab Republic of Egypt is embarking on a transformative path to build a New Republic grounded in knowledge and innovation. Given that technical and technological education lies at the core of Egypt Vision 2030—serving as the primary engine for economic growth and sustainable development—the Ministry of Education and Technical Education has placed strategic priority on developing this sector.

Within this framework, the Ministry›s vision for reforming the mathematics curricula in Applied Technology Schools stems from the conviction that strong foundational learning is essential for preparing a workforce equipped with 21st-century skills. These include critical thinking, innovative problem-solving, and logical algorithmic reasoning capable of translating national development goals into tangible industrial and technological advancements.

Driven by this vision and aligned with the Technical Baccalaureate framework, mathematics has been formulated to highlight its applied importance in accelerating technological breakthroughs. Artificial intelligence techniques and smart system simulations rely fundamentally on mathematical algorithms and principles.

Consequently, mastering mathematical concepts serves as the natural prerequisite for comprehending cutting-edge technological applications and engineering software solutions required by the future labor market. A thorough understanding of the mathematical content presented in this book directly enhances students› technical proficiency, driving industrial innovation and raising the global competitiveness of local products. Therefore, learning mathematics in Applied Technology Schools represents a strategic investment in human capital and Egypt›s digital future.

We wish you all success and guidance. May Egypt always remain at the forefront of nations through your dedicated efforts.

The Editorial Board

Unit One
(Ratio and Similarity)

Lesson 1: Ratio and Proportion and Their Properties.

Lesson 2: Direct Proportion (Variation).

Lesson 3: Inverse Proportion (Variation).

Lesson 4: Similarity of Polygons.

Lesson 5: Similarity of Triangles.





Lesson One:
Ratio, proportion and their properties

Learning Outcomes:

- Define the concepts of ratio and proportion.
- Calculate ratios between different quantities in real-life situations.
- Apply ratio and proportion and their properties to solve real-world problems.
- Accurately use geometric tools and measurement software.

Think & Discuss!



The ratio ensures that the new design maintains the same shape and accuracy as the original design.

Imagine you work in a metal fabrication factory. You are required to design a new part that has the same shape as the original part, but in a smaller size at a ratio of 1:2. What does this ratio mean?

If the length of the original part is 40 cm, what is the length of the new part?

Learn!



Definition:

A ratio is a comparison between two quantities of the same kind (such as length to length, or weight to weight), and both quantities must have the same units.

- In general, the ratio between the number x and the number y is written as:

$$\frac{x}{y} \quad \text{or} \quad x : y$$

- x is called the antecedent (or first term), y is called the consequent (or second term), and together, x and y are called the terms of the ratio.
- If both terms of a ratio are multiplied (or divided) by a non-zero constant, the value of the ratio does not change.

Example

- Find the ratio between the numbers 15 and 20 in simplest form.

Solution:

15: 20 = 3: 4

or

$$\frac{15}{20} = \frac{3}{4}$$

- Find the value of: x , If: $\frac{3}{7} = \frac{x}{21}$

Solution:

$3 \times 21 = x \times 7$

$$\therefore x = \frac{3 \times 21}{7} = 9$$

Example

If Ahmed's height is 160cm and his sister's height is 130 cm, find the ratio between Ahmed's height and his sister's height.

Solution:

The ratio between Ahmed's height and his sister's height:

=160 : 130 (÷ 10)

=16 : 13

Definition:

- **Proportion** is the equality of two or more ratios.
- In the proportion $\frac{a}{b} = \frac{c}{d}$:
  - a is called the **first proportional**,
  - b is called the **second proportional**,
  - c is called the **third proportional**, and
  - d is called the **fourth proportional**.
- Additionally, a and d are called the **extremes** of the proportion, while b and c are called the **means**.

Properties of Proportion:

- If $\frac{a}{b} = \frac{c}{d}$, then: ad = bc (The product of the extremes = The product of the means)

Example

In concrete manufacturing, if the ratio of sand to cement is 3:1, what is the quantity of sand required for 50 kg of cement?

Solution:

Let sand be a

and cement be b

a : b = 3 : 1

$\therefore \frac{3}{1} = \frac{a}{50}$

a = (3 × 50) ÷ 1 = 150 kg.

Example

Find the number that, when added to both terms of the ratio 7 : 11, makes it 3 : 4

Solution:

Let the number be: x

$$\frac{x+7}{x+11}=\frac{3}{4}$$

$$\therefore 4(x+7)=3(x+11)$$

$$4x+28=3x+33$$

$$4x-3x=33-28$$

$$\therefore x=5$$

Try to solve

1. Find the ratio between the two numbers 35 and 40 in its simplest form.
2. If the length of a metal rod is 80 cm and another is 40 cm, what is the ratio of the length of the first rod to the second?
3. Find the fourth proportional to the numbers 3, 7, and 6.

Check your understanding

How can understanding ratio and proportion help you accurately design a machine or an industrial product?

Exercises on Ratio, Proportion, and Their Properties

1. Find the number that, when added to the antecedent (numerator) of the ratio $\frac{29}{46}$ and subtracted from its consequent (denominator), gives the ratio $\frac{3}{2}$
2. Find the fourth proportional to the numbers:

   4, 12, 16, ….
3. Find the value of x that makes the following numbers proportional:

   3 + x , 5 +x , 8+x , 12+x
4. Find the number that, when added to the antecedent of the ratio $\frac{18}{30}$ and subtracted from its consequent, gives the ratio $\frac{5}{3}$
5. Find the fourth proportional to the numbers:

   6, 9, 12, …..
6. If the ratio between two numbers is 21 : 35, what is the number that, when added to the antecedent and subtracted from the consequent, makes the ratio 4 : 5?

Lesson Two:
Direct Variation

Learning Outcomes:

- Define the concept of direct variation (direct proportion).
- Connect direct variation to real-life and industrial situations, such as production or consumption.
- Solve mathematical problems using the direct variation formula.
- Apply direct variation to practical designs or calculations.

Think & Discuss!



Imagine that you work in a factory producing leather bags. If the number of workers doubles, the number of produced pieces also increases by the same ratio.

How can we describe this relationship?

Learn!



Definition:

Direct variation is a relationship between two variables such that when one increases, the other increases by the same ratio, and vice versa.

- It is said that y varies directly as x (written as $y \propto x$)

$$\text{If: } y = k\,x \qquad \text{or} \qquad \frac{y}{x} = k$$

Where k is the constant of variation and does not equal zero.

- If the variable x takes the values x_1 , x_2 and the variable y takes the values y_1 , y_2 respectively, then:

$$\frac{y_1}{y_2} = \frac{x_1}{x_2}$$

Example

If y varies directly as x , and y = 12 when x = 4 , find the constant of variation and the relation.

Solution:

Constant of variation:

$k = \frac{y}{x}$

$= \frac{12}{4} = 3$

Therefore, the relation is:

$y = 3x$

Example

In a garment factory, the cost of producing clothes varies directly as the number of pieces.

If the cost of producing 5 pieces of clothing is 100 pounds, what is the cost of producing 8 pieces?

Solution:

$$x_1 = 5 \ , \ x_2 = 8, \ \ y_1 = 100 \ , \ \ \ y_2 = ?$$

Since the relationship is directly proportional.

$\therefore \quad \frac{y_1}{y_2} = \frac{x_1}{x_2}$

$$\frac{100}{y_2} = \frac{5}{8}$$

$$y_2 = 100 \times 8 \div 5 = 160$$

Therefore, the cost of producing 8 pieces is 160 pounds.

Try to solve

1. In a factory, the production of 10 workers is 200 pieces daily. How many pieces do 15 workers produce?
2. On a farm, an irrigation machine pumps 60 liters of water in 3 hours. How many liters does it pump in 5 hours of irrigation?
3. If y = 2x , find the value of y when x = 9
4. If the distance covered by a car varies directly as time, what is the distance it covers in 3 hours if it covers 80 km in two hours?

Check your understanding

How can understanding direct variation help you calculate the production cost of a new product if the number of workers or working hours changes?

Exercises on Direct Variation

1. If y varies directly as x , and y = 15 when x = 5 , find the relation between them.
2. If a worker's wage for 4 hours of work is 200 pounds, what is the wage for 7 hours of work?
3. If the shadow length varies directly as the height of the pole, what is the shadow length of a 6-meter-high pole if the shadow of a 4-meter-high pole is 2 meters?
4. If $y \propto x$, and y = 14 when x = 42 , find:

   First: The relation between x and y

   Second: The value of y when x = 60

Lesson Three:
Inverse Variation

Learning Outcomes:

- Define the concept of inverse variation (inverse proportion).
- Connect inverse variation to real-life and industrial situations, such as completion speed and the number of workers.
- Solve mathematical problems using the inverse variation formula.
- Apply inverse variation to practical designs or calculations.

Think & Discuss!



Imagine that you work in a small workshop. If the number of workers increases, the time required to complete the work decreases by the same proportion.

How can we describe this relationship?

Learn!



Definition:

Inverse variation is a relationship between two variables such that if one increases, the other decreases by the same ratio, and vice versa.

- It is said that y varies inversely as x (written as $y \propto \frac{1}{x}$)

  If: $y=\frac{k}{x}$ or $x\,y=k$

Where k is the constant of inverse variation and does not equal zero.

- If the variable x takes the values x_1 , x_2 and the variable y takes the values y_1 , y_2 respectively, then:

$$\frac{y_1}{y_2} = \frac{x_2}{x_1}$$

Example

If y varies inversely as x, and y = 6 when x = 2, find the constant of variation and the equation.

Solution:

Constant of variation:

$k=x \times y$

$=2 \times 6=12$

Therefore, the relation is:

$x\ y=12$

Example

In a factory, the time required to produce a certain quantity of items varies inversely as the number of workers. If 12 workers take 10 hours, how many hours will 15 workers take?

Solution:

$x_1=12,\ \ y_1=10,\ \ x_2=15,\ \ y_2=?$

$$\frac{y_1}{y_2} = \frac{x_2}{x_1}$$

$$\frac{10}{y_2} = \frac{15}{12}$$

$$y_2 = 10 \times 12 \div 15 = 8$$

Therefore, the time = 8 hours.

Try to solve

1. If 5 workers complete a job in 12 hours, how many hours do 10 workers need to complete the same job?
2. If a car travels a certain distance in 2 hours at a speed of 60 km/h, what is the time if the speed is 80 km/h for the same distance?
3. If xy = 24, find the value of y when x = 6
4. If 4 workers complete a task in 18 hours, how many hours do 9 workers need to complete it?

Check your understanding

How can understanding inverse variation help you calculate the time required to complete a new project if the number of workers or machine speed changes?

Exercises on Inverse Variation

1. If y varies inversely as x , and y = 20 when x = 5 , find the equation relating them.
2. If the speed of completing a task is inversely proportional to time, and it was completed in 8 hours at a certain speed, what is the time if the speed is doubled?
3. If the pressure of a gas is inversely proportional to its volume, what is the pressure at a volume of 6 liters if the pressure at a volume of 4 liters is 12 units?
4. If $y \propto \frac{1}{x}$, and y = 3 when x = 2, find:

   First: The relation between x and y.

   Second: The value of y when x = 3

Lesson Four:
Similarity of Polygons

Learning Outcomes:

- Define the concept of similarity of polygons.
- Deduce the conditions for the similarity of polygons.
- Find the scale factor (similarity ratio).
- Deduce the relationship between the perimeters of two similar polygons.
- Deduce the relationship between the areas of two similar polygons.
- Apply polygon similarity to design scale models of mechanical parts or architectural structures.
- Apply polygon similarity to solve real-world problems.

Think & Discuss!

The following figure shows:

Wind turbines in the Red Sea Governorate for generating eco-friendly renewable energy.



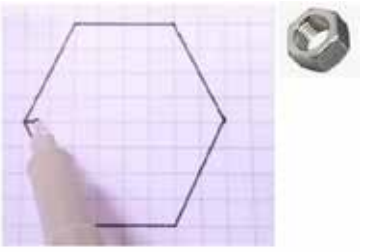
If you create a scale model of a critical mechanical gear inside the turbine where the gear's actual diameter is 2000 mm (2 m)

and while transferring the dimensions to make the scale model according to a certain scale, you made a calculation error of only 1% (a deviation of 20 mm in actual dimensions), whereas the required quality standards demand an accuracy within 0.1% (a deviation not exceeding 2 mm), what would you do?



Learn!

Similarity of Polygons:



Here is a model of a hexagonal nut:

- Find the measures of the corresponding angles in the original piece and the model. What do you notice?

You will notice that: "The angle measures in both the original shape and the model are 120°."

- Find the ratio between the lengths of corresponding sides in the original piece and the model. What do you notice?

You will notice that: "The ratio between the lengths of corresponding sides is always constant. This constant ratio is called the similarity ratio or scale factor.

Definition:

"Two polygons with the same number of sides are similar if their corresponding angles are congruent, and their corresponding side lengths are proportional."

The Fundamental Law of Similarity:

If we have an original geometric polygon and a scaled-down polygon of it, the two conditions for similarity are:

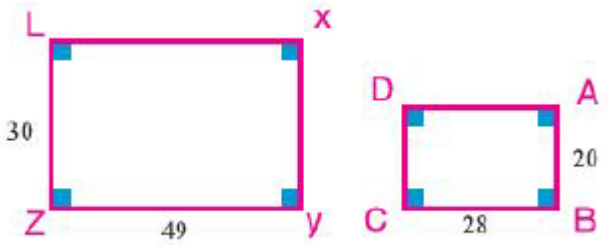
1. Corresponding angles are congruent (equal in measure).
2. Corresponding sides are proportional.

We use the symbol (~) to denote the similarity of two polygons. The corresponding vertices must be written in matching order to easily write the proportions between corresponding sides.

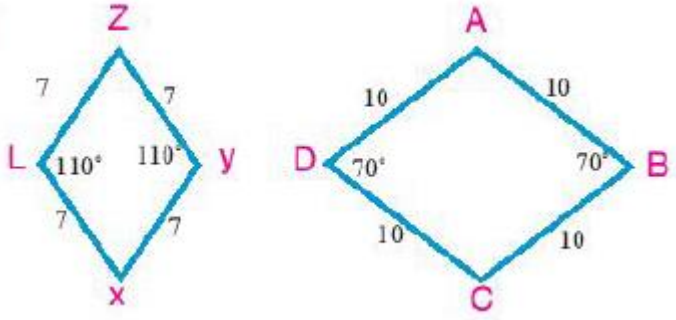
Example

Show which of the following pairs of polygons are similar. Write similar polygons in order of their corresponding vertices and determine the similarity ratio (lengths are given in centimeters).

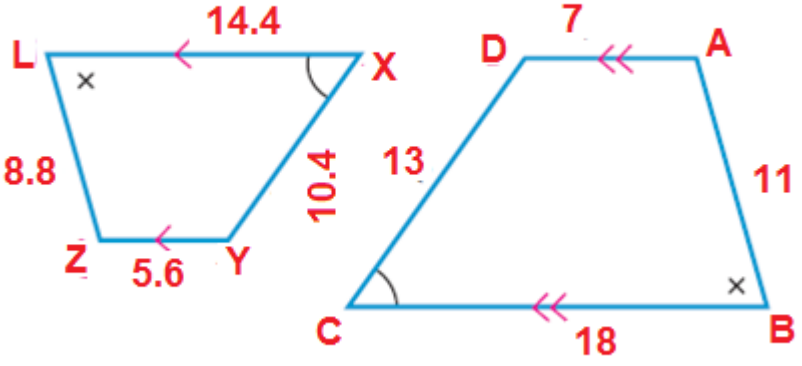
A



B



C



Solution:

a) The corresponding sides are not proportional

$\Rightarrow$The two polygons are not similar.

b) The corresponding sides are proportional

$\Rightarrow$The two polygons are similar.

Polygon ABCD $\sim$ Polygon LXYZ

Scale factor = $\frac{10}{7}$

c) The corresponding sides are proportional

$\Rightarrow$The two polygons are similar.

Or use parallelism to prove that the two polygons are similar as follows:

"Supplements of equal angles are equal in measure."

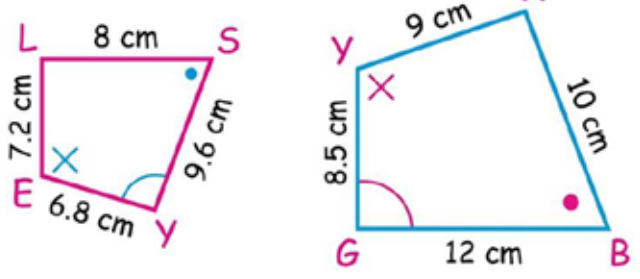
Polygon BCDA ~Polygon LXYZ

Scale factor = $\frac{5}{4}$

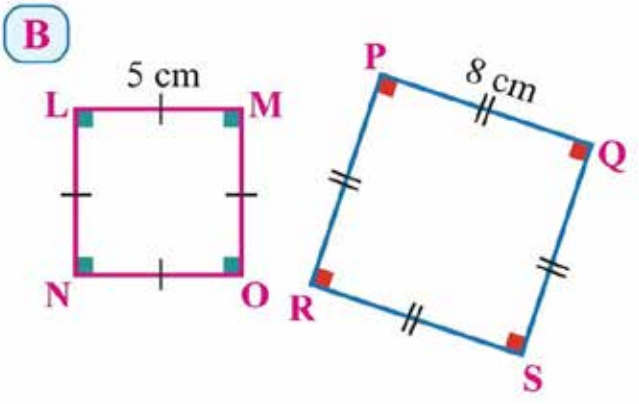
Try to solve

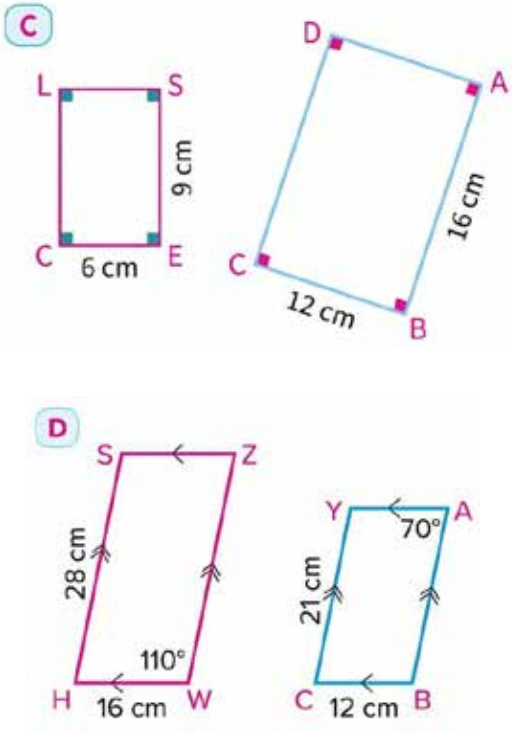
Show which of the following pairs of polygons are similar. Write similar polygons in order of their corresponding vertices and determine the similarity ratio.

A



B





Note (1):

- One of the polygons can be an enlargement of the other if the scale factor is greater than one ($k > 1$).
- One of the polygons can be a reduction of the other if the scale factor is greater than zero and less than one ($0 < k < 1$).
- The two polygons can be congruent if the scale factor is equal to one ($k = 1$).

Note (2):

All regular polygons with the same number of sides are similar.

Think

- Are all squares similar?
- Are all rhombuses similar?
- Are all rectangles similar?
- Are all parallelograms similar?

Relation between two perimeters of similar polygons

If two polygons are similar, then:

$$\frac{\text{Perimeter of first polygon}}{\text{Perimeter of second polygon}} = \text{The scale factor}$$

Relation between two areas of similar polygons

If two polygons are similar, then:

$$\frac{\text{Area of first polygon}}{\text{Area of second polygon}} = \text{Square of the scale factor}$$

Example

In the opposite figure:

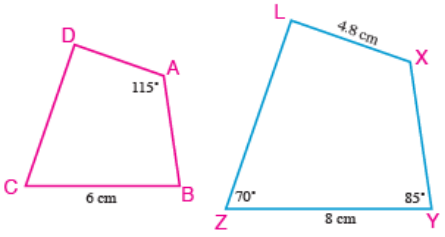
Polygon ABCD ~ Polygon XYZL

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(1) Find m(∠XLZ), and the length of $\overline{AD}$

(2) If the perimeter of Polygon ABCD = 19.5cm , find the perimeter of Polygon XYZL.

(3) The ratio between the area of Polygon ABCD and the area of Polygon XYZL.



Solution:

1) Polygon ABCD ~ Polygon XYZL

The corresponding angles are congruent , The corresponding sides are proportional.

M(∠XLZ)= 90°

Scale factor = $\frac{3}{4}$

$\frac{3}{4} = \frac{AD}{4.8}$ Ad =3.6 cm

2) $\frac{19.5}{\text{Perimeter of polygon XYZL}} = \frac{3}{4}$

Perimeter of polygon XYZL = 26 cm

3) $\frac{\text{Area of first polygon}}{\text{Area of second polygon}} = \frac{9}{16}$

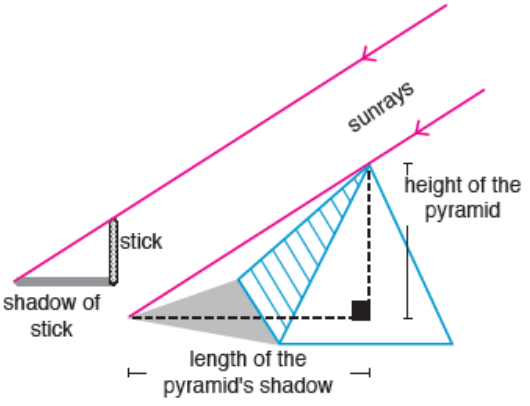
Try to Solve

ABCD is a rectangle in which AB = 5 cm and BC = 7 cm. Find the dimensions of another rectangle similar to it if:

1. The scale factor is 1.5
2. The scale factor is 0.5

Check Your Understanding

Using a stick of length 1.5 m, and based on what you have learned about similarity, find the height of the Pyramid of Khufu. (Note: The Greek mathematician Thales measured the height of the Pyramid of Khufu using this same method).



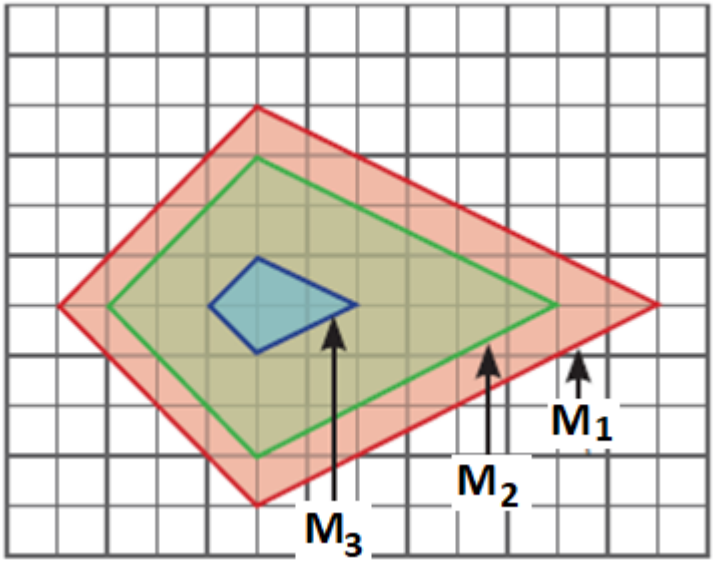
Exercises on Similarity of Polygons

1. If Polygon ABCD ~ Polygon XYZL, complete:

1) $\frac{AB}{BC} = \frac{\ldots\ldots}{YZ}$ 2) AB × ZL = XY × …….

3) $\frac{BC-YZ}{YZ} = \frac{\ldots\ldots-LX}{XL}$ 4) $\frac{P.of\ the\ polygon\ldots\ldots\ldots\ldots}{P.of\ the\ polygon\ldots\ldots\ldots\ldots} = \frac{XY}{AB}$

2. In the following figure, Polygon M_1 ~M_2 ~M_3. Find the scale factor for each polygon.



3. Polygon ABCD ~ Polygon XYZL. If AB = 32 cm, BC = 40 cm, XY = 3m - 1, and YZ = 3m + 1, find the numerical value of m.

4. A golden-rectangle-shaped box has a length of 16.2 cm.
Calculate the width of the box to the nearest centimeter.
Note: A Golden Rectangle is a rectangle that can be divided into a square and another rectangle similar to the original. The ratio of length to width is the Golden Ratio, approximately 1.618.

5. A rectangular sports hall has dimensions of 8 m and 12 m. Its floor was covered with wood at a cost of 32,000 EGP. Calculate using similarity the cost of covering the floor of a larger rectangular hall using the same materials if its dimensions are 14 mand 21 m.

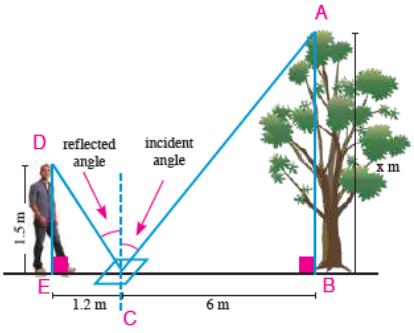
Lesson 5: Similarity of Triangles

Learning Outcomes:

- Understand the fundamental concept of triangle similarity.
- Deduce the specific criteria and cases for similarity.
- Apply similarity theorems to solve complex mathematical and real-life problems

Think and Discuss

A person wanted to determine the height of a tree. He placed a mirror on the ground 6 meters from the base of the tree. He then walked backward until the top of the tree was visible in the center of the mirror. At that point, he was 1.2 meters away from the mirror. If the person's height is 1.5 meters.



- How can similarity be used to find the height of the tree?
- Note: Angle of incidence = Angle of reflection.

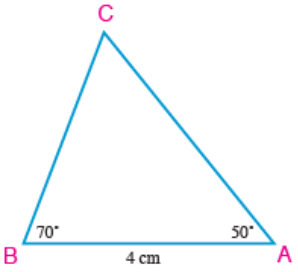
Learn:

- Draw Δ ABC in whcich m (∠A) = 50° , m (∠B) = 70° , AB = 4 cm
- Draw Δ XYZ in whcich m (∠X) = 70° , m (∠Y) = 70° , XY= 6 cm

Find to the nearest AC , BC , XZ , YZ

Use the calculator to find the following ratios:

$$\frac{AC}{XY}, \frac{BC}{YZ}, \frac{ZB}{XZ}$$



What do you notice:

- Are the ratios equal?
- Are the corresponding angles congruent (equal in measure)?

- Are the corresponding sides proportional?
- What do you conclude about these two triangles?

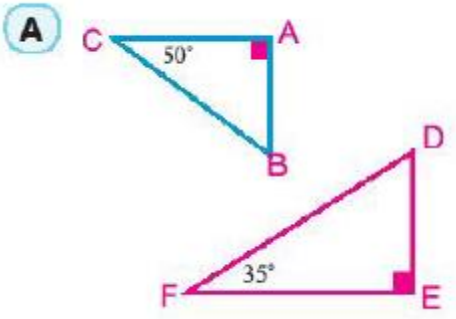
Defining Similarity of triangles

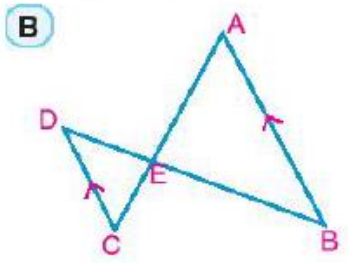
Two triangles are similar if only one of the following conditions is satisfied:

1. Corresponding angles are congruent (equal in measure).
2. The lengths of corresponding sides are proportional.

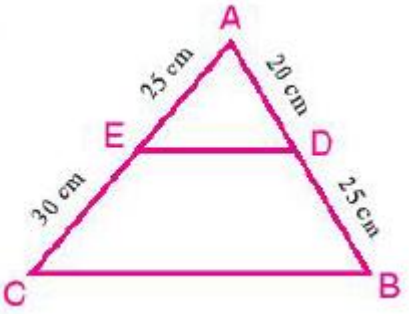
Example:

Which of the following pairs of triangles are similar? Write the similar triangles in order of corresponding vertices:





C



Solution:

a) Some corresponding angles are congruent and some are not congruent, then the two triangles are not similar.

b) The corresponding angles are congruent, and then the two triangles are similar.

c) The corresponding sides are not proportional, and then the two triangles are not similar.

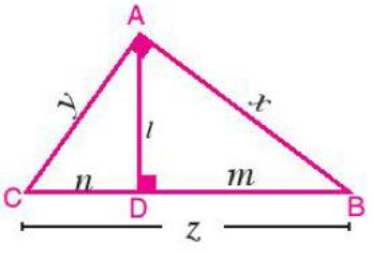
Corollary:

If a perpendicular is drawn from the vertex of the right angle of a right-angled triangle to the hypotenuse, the triangle is divided into two similar triangles; both of them are similar to the original triangle.

Try to Solve

In the opposite figure, Triangle ABC is right-angled at A, and $\overline{AD} \perp \overline{BC}$.

1) Complete: Δ ……. ~Δ ……. ~ Δ …….



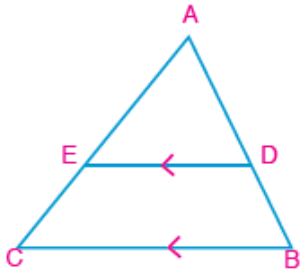
2) If x, y, z, L, m, n are the lengths of segments, complete the proportions:

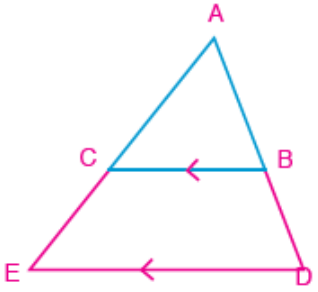
$$\frac{X}{Z}=\frac{M}{.....}\ ,\ \frac{X}{Z}=\frac{L}{.....}\ ,\ \frac{M}{L}=\frac{X}{.....}\ ,\ \frac{L}{...}=\frac{...}{L.}$$

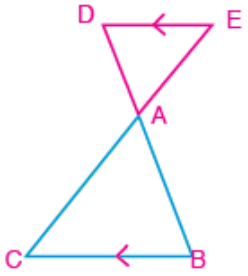
$$\frac{X}{.....}=\frac{.....}{X.}\ ,\ \frac{.....}{Y}=\frac{Y}{.....}\ ,\ \frac{L}{X}=\frac{.....}{Z}\ ,\ \frac{L}{X}=\frac{...}{Y}$$

Theorem (1):
If a line is drawn parallel to one side of a triangle intersecting the other two sides , the resulting triangle is similar to the original triangle.

In the following figures:







If $\overleftrightarrow{DE}$ // $\overleftrightarrow{BC}$ then Δ ADE ~ Δ ABC.

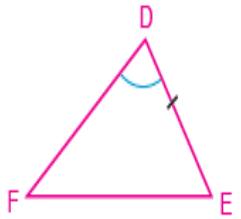
Theorem (2):
If an angle of one triangle is congruent to an angle of another triangle and the sides including these angles are proportional, the triangles are similar.

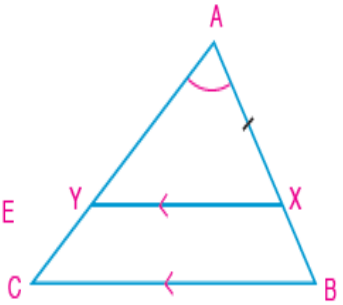
In the opposite figure**:**

Givens: ∠A ≡ ∠D

$$\frac{\boldsymbol{AB}}{\mathbf{DE}}=\frac{\boldsymbol{AC}}{\mathbf{DO}}$$

Rquired:
Δ ABC ~ ΔEDO





Proof:

Mark the point X on $\overline{AB}$ where AX = DE, then draw $\overrightarrow{XY}$ // $\overline{BC}$ and cuts $\overline{AC}$ in Y.

Where $\overrightarrow{XY}$ // $\overline{BC}$, then Δ ABC ~ Δ AXY (the opposite aangles are congruent).

Then: $\frac{AB}{AX} = \frac{AC}{AY}$

$\frac{AB}{DE} = \frac{AC}{DO}$, AC = DE

$\frac{AB}{AX} = \frac{AC}{DO}$, AY = DO

Δ AXY ≡ ΔDEO (S.S.A case)

Then, Δ AXY ~ ΔDEO

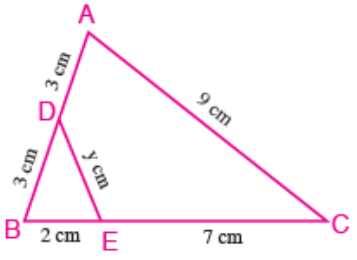
We deduce that:Δ ABC ~ ΔDEO

Example:

In th opp. Figure the two triangles BDE and BCA, find the value of Y where:

∠B is a comman angle.

$\frac{BD}{BC} = \frac{BE}{BC} = \frac{1}{3}$ **(check that)**



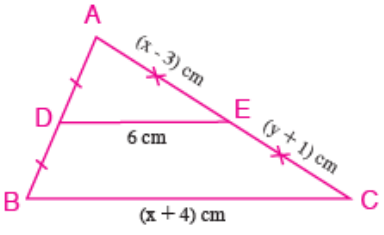
Then Δ BDE~ ΔBCA and we deduce that:

$\frac{DE}{CA} = \frac{1}{3}$

$\frac{Y}{9} = \frac{1}{3}$ **, Y = 3 cm**

Try to Solve

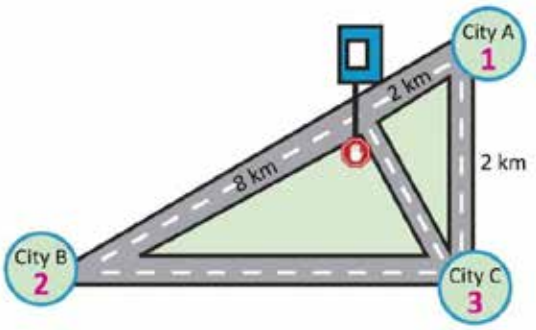
In the opposite figure, find the value of X and Y:



Check Your Understanding

Check Your Understanding

A service and fueling station is to be constructed on a highway at the intersection of a side road leading to City C. The side road is perpendicular to the highway between Cities A and B.



- How far should the station be from City C?
- What is the distance between Cities B and C?

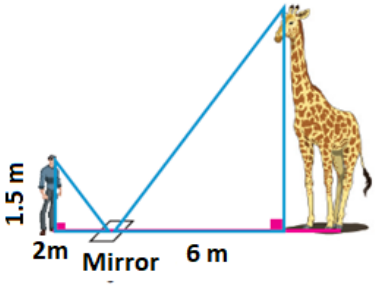
Exercises on Similarity of Triangles

1. Group A and group B express the lengths of sides of different triangles in cm. Write in front of each triangle from group (A) the letter of its similar triangle in group (B):

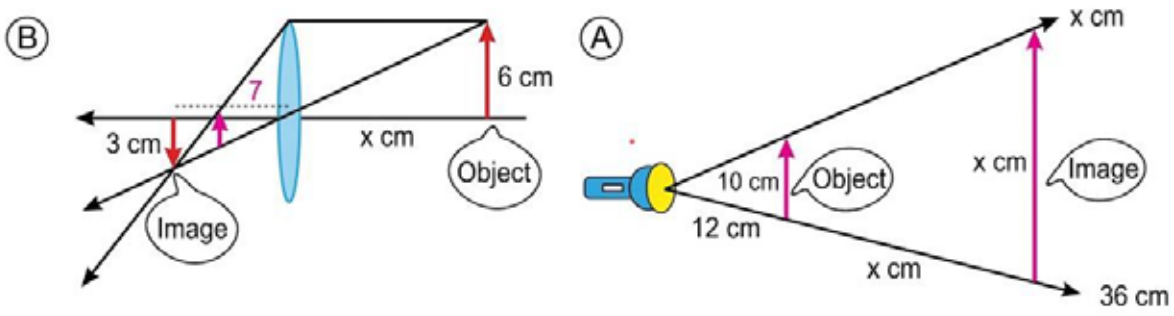
Group (B)	
5 , 4 , 2.5	a
14 , 13.5 , 8	b
55 , 35 , 25	c
11 , 11 , 11	d
6 , 4 , 3.5	e
10 , 6 , 8	f
42 , 54 , 32	g

Group (A)	
6 , 6 , 6	1
11 , 7 , 5	2
10 , 8 , 5	3
12 , 8 , 7	4
28 , 27 , 16	5

2. Ahmed wanted to find the height of a giraffe at the zoo. He placed a mirror on the ground 2 meters far from him and 6 meters from the giraffe. If Ahmed's height is 1.5 meters, calculate the height of the giraffe.



3. Calculate the scale factor, find the value of X in each of the following:


B
7
6 cm
3 cm
x cm
Object
Image
A
x cm
10 cm
Object
x cm
Image
12 cm
x cm
36 cm

Unit Project

In your specialization, use what you have learned about complex numbers and equations to prepare a project according to the following specifications.

Project Assessment Form

School: .....

Project Title: .....

Date: .... / .... / .....

Project Participants: .....

Project Rubric

M	Criterion	Beginning	Intermediate	Advanced	Proficient	Score
1	**Problem Identification and Overall Project Treatment**	Attempts are made, but the problem does not appear to be clear.	The minimum required for understanding the problem appears, without sufficient evidence.	Demonstrates understanding of the topic, but the evidence is not clear or sufficient.	Demonstrates a correct and clear understanding of the problem and presents multiple pieces of evidence.	
2	**Availability of Creative Solutions to the Problem**	Is unable to generate strategies to solve the problem.	Begins but fails to complete the solution.	Begins in a reasonable way but leaves some parts of the task incomplete.	Uses a clear and organized approach and demonstrates high-level creative skills.	
3	**Students' Presentation Skills**	Needs guidance in organizing the	Organization is weak and the sequence does not	Organization or logical sequencing is present, but	Includes a clear introduction, main topic,	

M	Criterion	Beginning	Intermediate	Advanced	Proficient	Score
		presentation. The points are presented randomly, making them difficult to follow.	facilitate understanding.	the presentation does not always follow a clear structure.	and conclusion. The points are presented in a logical sequence using appropriate language.	
	Final Score					

Committee Decision:

.....

.....

.....

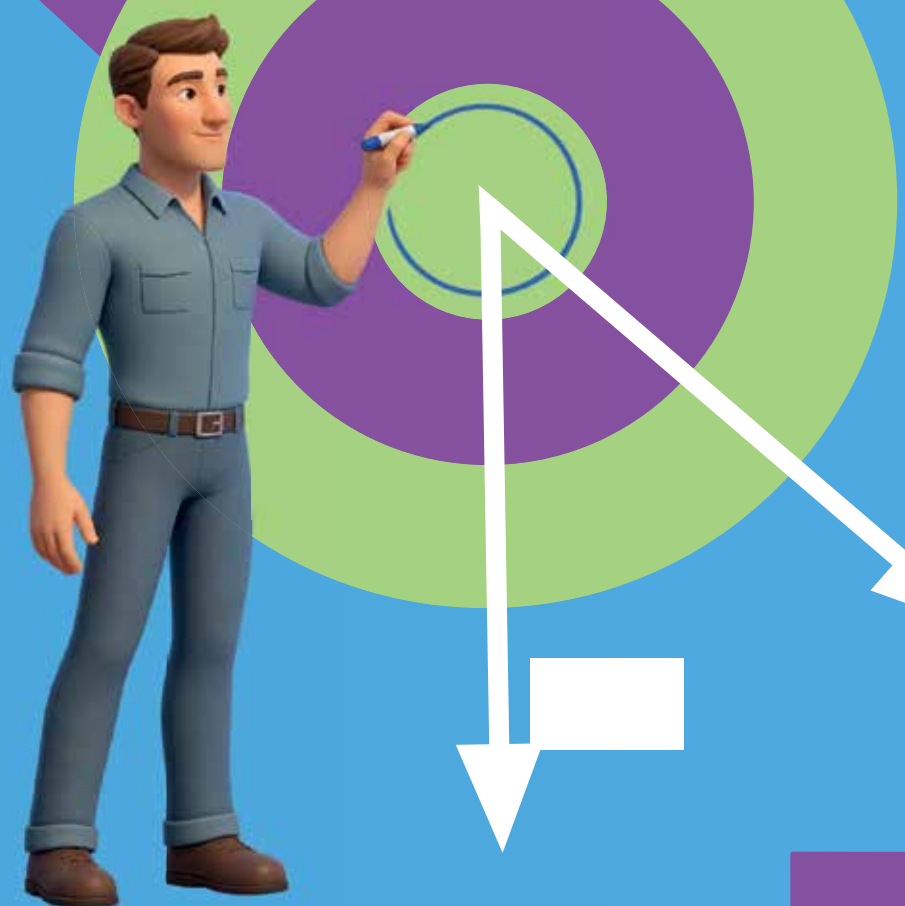
Signature: .....

Unit Two
Degree measure and radian measure of an angle

Lesson 1: Directed Angle

Lesson 2: Degree measure and radian measure of an angle

Lesson 3: Circular Sector and Circular Segment



Lesson (1)
Directed angle

Learning Outcomes:

- Define the concept of directed angle.
- Determine the standard position of a directed angle.
- Determine the positive and negative measures of a directed angle.
- Show the position of a directed angle in the coordinate plane.
- Deduce equivalent angles.

Think and Discuss

Robot Arm: On a production line, two engineers programmed a robot arm to move at an angle of 60° to transfer a metal part from position (A) to position (B). However:
First Engineer: The arm moved and reached the part successfully.
Second Engineer: The arm moved in the exact opposite direction and hit the safety wall.
What was the cause of this error?

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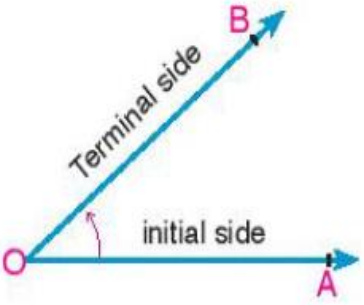


Learn

Deffinition:

A directed angle is an ordered pair of two rays that form the sides of the angle, having a common starting point which is the vertex of the angle.

- The opposite figure represents the directed angle $\angle(AOB)$



- The two sides of the directed angle are expressed by the ordered pair $(\overrightarrow{OA}, \overrightarrow{OB})$, where $\overrightarrow{OA}$ is the initial side , $\overrightarrow{OB}$ is the terminal side for the angle whose vertex is the point O.

Critical thinking

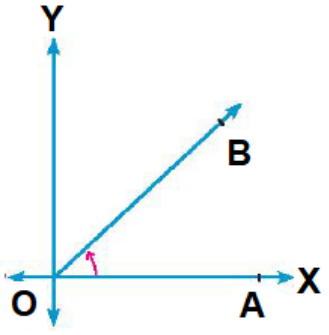
- Is $(\overrightarrow{OA}, \overrightarrow{OB}) = (\overrightarrow{OB}, \overrightarrow{OA})$? Why?
- Is the directed angle AOB is the same as the directed angle BOA? Why?

Standard position of the directed angle

The standard position of a directed angle is fundamental for programming robot trajectories and guiding computer numerical control systems. By placing the vertex at the origin of the coordinate system and referencing all rotation from the positive x-axis, engineers ensure precise machine movement and a unified language for engineering measurements.

A directed angle is in standard position if its vertex is at the origin of a Cartesian coordinate system, and its initial side lies on the positive x-axis.

In the opposite figure, AOB is a directed angle. Why?



Example:

Which of the following ordered pairs represents a directed angle in standard position? why?

A $(\overrightarrow{CA}, \overrightarrow{CD})$

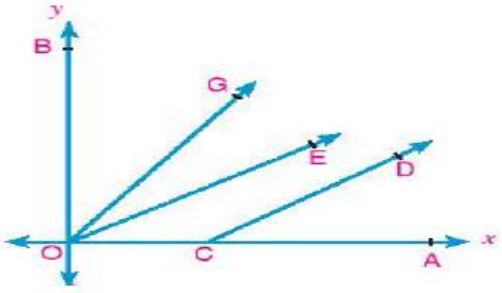
B $(\overrightarrow{OA}, \overrightarrow{OE})$

C $(\overrightarrow{OE}, \overrightarrow{OA})$

D $(\overrightarrow{OA}, \overrightarrow{OG})$

E $(\overrightarrow{OB}, \overrightarrow{OG})$

F $(\overrightarrow{OA}, \overrightarrow{OB})$



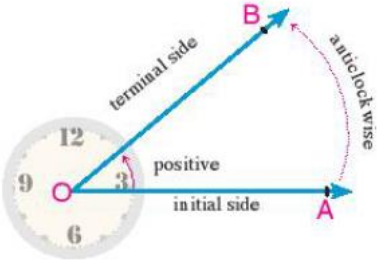
Answer:

A) It does not represent a directed angle in standard position because the vertex of the angle does not lie on point O (the origin).

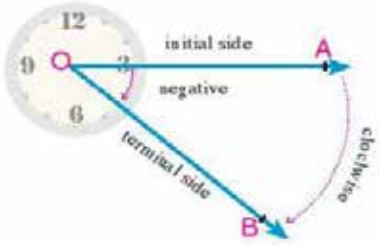
(Complete Answer with Full Geometric Explanation)

Positive and negative measure of a directed angle

Positive Measure: the measure of a directed angle is positive when the direction of rotation from the initial side to the terminal side is anticlockwise.



Negative Measure: the measure of a directed angle is positive when the direction of rotation from the initial side to the terminal side is clockwise.



Rule:

For any directed angle, the sum of the absolute values of its positive and negative measures equals 360˚, This means that if the positive measure is $\theta°$, then its corresponding negative measure is $\boldsymbol{\theta°}$ **- 360°**

Example:

Find the measure of the angle indicated by $\boldsymbol{\theta}$ in each of the following:



Answer:

A) The direction of the directed angle is clockwise, therefore:

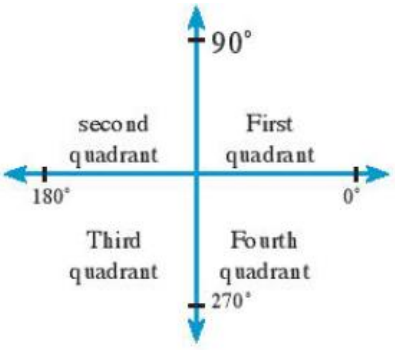
$$m(\angle\theta) = -(360° - 55°) = -305°$$

B) The direction of the directed angle is anticlockwise, therefore:

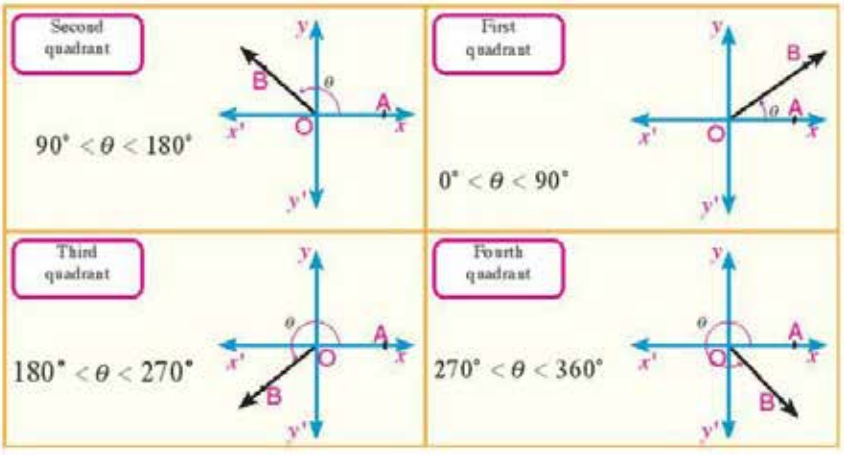
$$m(\angle\theta) = 360° - 33° = 327°$$

Position of a directed angle in the coordinate

The coordinate plane is divided into 4 quadrants as the opposite figure:



If ∠AOB is a directed angle in standard position with a positive measure θ, then its terminal side $\overrightarrow{OB}$ can lie in one of the quadrants as shown in the following figure:



If the terminal side lies on one of the coordinate axes, the angle is called a **Quadrantal Angle**. Thus, angles measuring 0° , 90°, 180°, 270° and 360° are quadrantal angles.

Example:

Determine the quadrant in which each of the following angles lies:

A 48° B 217° C 135° D 295° E 270°

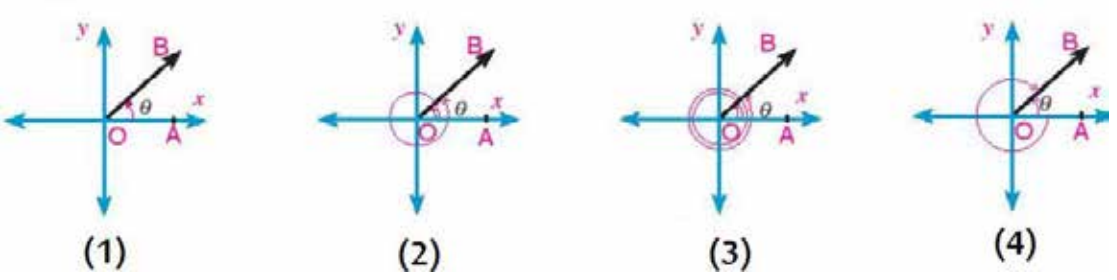
Answer:

A $0° < 48° < 90°$ then it lies in the first quadrant.
B $180° < 217° < 270°$ then it lies in the third quadrant.
C $90° < 135° < 180°$ then it lies in the second quadrant.
D $270° < 295° < 360°$ then it lies in the fourth quadrant.
E 270° is a quadrantal angle.

Coterminal angles

- Coterminal Angles: Are directed angles in standard position that all have the same terminal side.
- To find coterminal angles with an angle of measuring θ, we add or subtract a full rotation (360°) or a multiple of it

$$\theta \pm n \times 360° \ , n \in z^{+}$$



Example:



In a smart irrigation system, a sprinkler arm rotates around its pivot point by a directed angle with a positive measure θ to cover the largest possible watering area. If we want to rotate the sprinkler arm in the opposite direction while reaching the exact same final line:

1. Calculate the angle measure in which the arm must rotate.

2. If the sprinkler breaks down after completing 2 full rotations in its original direction, find the coterminal angle measure at which it stopped working.

Answer:

1. The angle measure in which the arm must rotate:

$$120° - 360° = -240°$$

2. The coterminal angle measure at which it stopped working:

$$120° + 2 \times 360° = 840°$$

Try to solve

Find two coterminal angles— one positive and one negative—for each of the following angles:

a) 40° b) 150° c) -240° d) -180°

Check your understanding

1. Determine the quadrant in which each of the following angles lies:

a) 55° b) 245 ° c) 570°

2. Detremie the negative measure for each of the following:

a) 90° b) 214° c) 330°

3. Find the least positive measure for each of the following:

a) -55° b) -220° c) 450° d) 900°

Exercises on the directed angle

1. Choose:

a. The directed angle whose sides $(\overrightarrow{OP}, \overrightarrow{OQ})$its initial side is…

a) $\overrightarrow{OP}$ b) $\overrightarrow{OQ}$ c) O d) $\overrightarrow{PQ}$

b. If the directed angle is in the standered positin, then its vertex lies on …….

a) x-axis

b) y-axis

c) the origin

d) one of the four quadrants.

c. Angle of measure 220° les on the …….. quadrant.

a) first b) second c) third d) fourth

d. Which of the following angles equivalent to the angle of measure 60° ?

(a - 60°

(b 420°

(c 300°

(d -120°

2. Find the positive measure for:

-300° , -90° , -125°

3. **Technology**:

In a digital control system for a lathe machine, the workpiece rotated by -7103. Find the smallest positive coterminal angle measure for this rotation, then determine the quadrant in which the tool stops.



Lesson (2)
Degree measure and radian measure of angle

Learning Outcomes:

- Define the angle and its methods of measurement.
- Differentiate between degree measure and radian measurements.
- Recall the relationship between degrees and radians, and convert between them.
- Apply the concept of radians in industrial contexts.
- Relate angle measurement to the movement of gears and power transmissions.
- Solve problems related to the rotation of mechanical parts.

Introduction

In applied technology fields—particularly in mechanical, industrial control, automotive, and machine manufacturing specializations—angles are essential measurements relied upon for system operations.
When designing a transmission system or adjusting a gear within a machine, technicians and engineers must precisely know the rotation amount of component parts. Even a minor error in a gear's rotation angle can lead to:

Irregular motion

Increased friction

Loss of transmitted power

Damage to machine components

Therefore, various angle measurement systems are utilized, most notably:

Degree measure.

radian measure.

Think and discuss



In a smart factory, an industrial robotic arm works on assembling metal parts. The arm rotates around a fixed point, and the control system must know the exact amount of rotation.How do we describe the arm's amount of rotation? Can we simply use terms like "large rotation" or "small rotation"?

Why do machines require precise angle measurements?

Why do certain applications use radians instead of degrees?

Learn

Concept of an Angle

When an object rotates around a point or an axis, the amount of this rotation is called an angle.

In other words: an angle is the amount of rotation or divergence between two rays or directions that share a common point called the vertex.

Angles appear in:

Gear movement

Motor rotation

Robotic arms

Industrial valves

First: Degree measure for an angle

Definition: It is a measurement system based on dividing a full rotation into 360 equal parts.

Each part is called a **degree (1°)**.

A degree is divided into: **60 minutes (60')**.

A minute is divided into: **60 seconds (60")**.

Example(1)

If a gear rotates by an angle of 45°, what does that mean?

This means it rotates by 45 parts out of 360 parts of a full rotation, which is $\frac{1}{8}$ of a rotation

Example(2)

Convert 30° 20' into minutes

Answer

30° = 30 × 60 = 1800'

Then, 20 + 1800 = 1820'

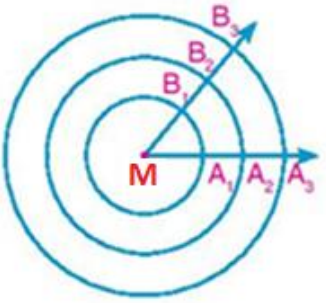
Try to solve

In a machine, a rotating shaft moved through an angle of 90° before stopping. What is the ratio of the shaft's rotation compared to a full rotation?

1. Did the machine move a quarter turn or a half turn?
2. Where can we see a similar movement in industrial applications?

Second: Radian Measure

Group work



1- Draw a set of concentric circles.

2- Find the ratio of the length of the arc and the length of the radius of its corresponding circle – what do you notice?

We notice that the ratio of the length of the arc of any central angle, and the radius of its corresponding circle equals constant quantity.

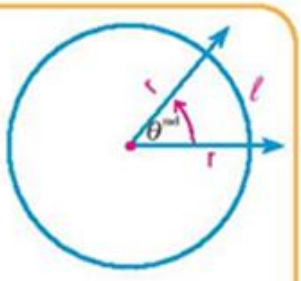
$$\text{i.e.: } \frac{\text{length of } \overset{\frown}{A_1B_1}}{MA_1} = \frac{\text{length of } \overset{\frown}{A_2B_2}}{MA_2} = \frac{\text{length of } \overset{\frown}{A_3B_3}}{MA_3} = \text{constant quantity.}$$

and this constant is the radian measure of the angle. The radian measure of the central angle (θ^{rad}). $= \frac{\text{length of the arc which the central angle subtends}}{\text{Radius of this circle}}$

Definition

If θ^{rad} is the radian measure of the central angle in a circle of radian r subtends an arc of length ℓ, then length of the arc equals the product of the radian measure of the central angle and the radius of its circle:

$$\ell = \theta^{rad} \times r$$



We deduce that:

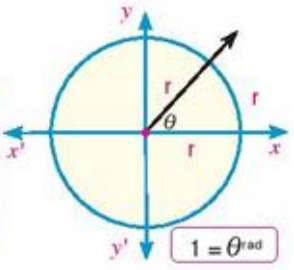
Radian is a unit of angle measurement that depends on the arc length resulting from the rotation of the radius.

When: Arc length = Circle radius, the resulting angle is called: 1 radian.

Definition

Radian angle

It is a central angle in a circle subtends an arc of length equals the radius of this circle.



We find that the degree and circular measures are two methods for expressing angle measures. The degree measure divides the circle into 360 degrees, while the circular measure measures the angle in radian measure in terms of arc length and radius.

Rule of radian measure

If the length of an arc is l and its radius is r, then: $\theta = \frac{l}{r}$

Why we use radian measure in technology?

Because radian depends on the ration $\frac{\text{length}}{\text{length}}$

So, it is a real number can be used directly in:

- Physics equations

- Programming

- Control systems

- Robotics

Relation between degree measure and radian measure

You know that the measure of a central angle in a circle is equal to the measure of its intercepted arc.

That is:

A central angle with a degree measure of 360° corresponds to an arc length of **2πr**

In a unit circle:

2π radians in circular measure is equivalent **360°** in degree measure.

That is:

π radians is equivalent to 180 °

If we have an angle with a circular measure of θ radians and a degree measure of x degrees, then:

$$\frac{x^\circ}{180^\circ} = \frac{\theta^{rad}}{\pi}$$

Therefore, conversion between degree measure and circular measure can be done using the previous relationship.

Note: In a unit circle, the length of the radius is equal to 1; for this reason, it is called a unit circle.

Rule of conversion

From degree to radian:

$$\theta(rad) = X^\circ \times \frac{\pi}{180}$$

From radian to degree:

$$X^\circ = \theta(rad) \times \frac{180}{\pi}$$

Example(1): Geometry application

Find the measure in radians. If an indicator in a measuring device moves through an angle of: $75^\circ 30'$

Answer:

$$30' = \frac{30}{60} = 0.5^\circ$$

therefore the degree measure is 75.5°

and the radian measure is $75.5 \times \frac{\pi}{180}$

$$= 1.318\ rad$$

Check your understanding

In a motor rotating shaft, a deviation occurred amounting to:

$2.15rad$

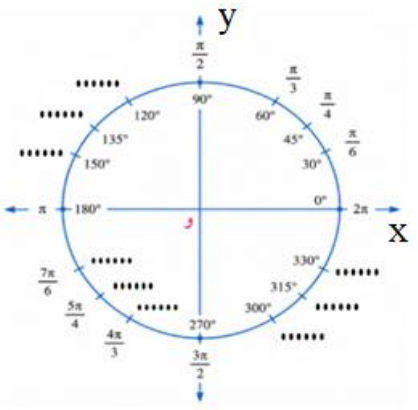
Find:

The measure in degrees

The measure in degrees , minutes and seconds

Try to solve

The opposite figure represents some special angles are written in radian (outside the circle) and in degrees (inside the circle). Complete the figure by writing the measure of each angle:



Example (2): Mechanic application

A gear with a radius of $10cm$ rotated through an angle of $2.5\ rad$
Find the length of the arc.

Answer:

$$l = \theta r$$
$$l = 2.5 \times 10$$

Then, $l = 25cm$

Check your understanding

A hydraulic arm moves along a circular path with a length of $120cm$ and the movement angle is $135°$. Find the length of its radius.

Example (3): Industrial Problem

A robot on a production line needs to rotate $90°$

However, the internal control system operates in radians.

Discuss: Why is the angle converted to $\frac{\pi}{2} rad$

before executing the movement?

Exercises on degree and radian measure

1. Choose:
   1- Full rotation = …..
   a) $90°$ b) $180°$ c) $360°$ d) π

2- 180° equals……

a) π/2 b) π c) 2π d) π/4

3. if the radius length is increased , while the angle measure is constant, then the arc length is….

a) doubled b) decreased c) not changing

2. Convert to degrees:

60° to radian , 2π

3. Complete:

1. 360° =.....radian
2. π radian = …. Degrees
3. Angles are used in the industry to measure…….
4. the gear rotation depends on ….. movement.
5. the whole circle equals …. Degrees, …. Radian

4. select an industrial machine containing rotating parts, such as:

- Engine / Motor
- Transmission / Gearbox
- Industrial Robot

Then write:

- Where do angles appear?
- What is the importance of measuring them?
- Can they be expressed in both degrees and radians?

5. Industrial Automation

Small pulley $r_1 = 5cm$ and a big bully $r_2 = 15cm$

If the small is rotated $3rad$, find:

- Belt movement length.
- Rotation angle of the large pulley.

Then discuss: How does the radius affect the rotation angle?

6. Robotics and AI Application An industrial camera scans an angle of: $\frac{5\pi}{6}rad$,

Find:

- The measure in degrees.
- The arc length if the radius is: $18m$

Lesson(3)
Circular Sector and Circular Segment

Learning Outcomes

- Distinguish between the concepts of a circular sector and a circular segment in geometric figures.
- Deduce and calculate the areas of circular sectors and circular segments using both circular measure (radians) and degree measure .
- Apply mathematical laws to solve real-world industrial problems, such as calculating the motion paths of robotic arms, designing gear systems, and estimating material waste in cutting processes.

Think and discuss

In the world of applied technology, mechanical engineering, and industrial control, we rarely deal with complete circles alone. From radar screens to the movement of car engines, windshield wipers, and the reach of an industrial robotic arm—all rely on parts of a circle. Understanding how to calculate the areas of these parts (circular sectors and circular segments) ensures that engineers and technicians achieve design precision, material efficiency, and optimized mechanical performance for machinery.

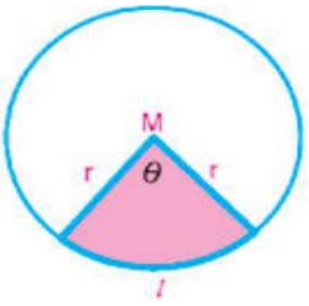
Industry

Imagine you are a programming technician for an automatic painting machine in a car factory. The machine arm moves from its pivot point in a circular path, scanning a specific angle to paint a designated fan-shaped section of a car door.



How can you accurately calculate the area of this region to adjust the amount of paint pumped by the machine, ensuring no material is wasted or left unpainted?

Learn



Definition

The cicular sector:

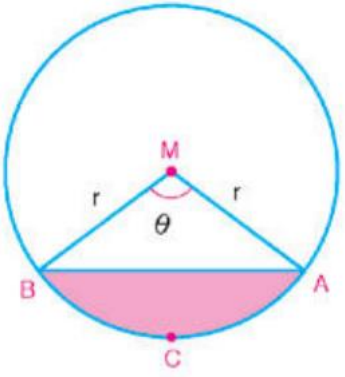
It is a part of the circle's surface bounded by two radii and an arc (resembling a slice of pizza).

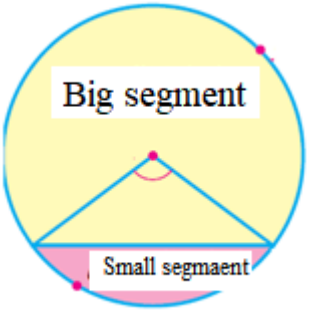
Where the key variables we will use are:

- **r**: Length of the circle's radius.
- **L**: Arc length.
- $\boldsymbol{\theta}^{\mathbf{rad}}$ **:** Central angle in circular measure (radians).
- **θ ° :**Central angle in sexagesimal measure (degrees)

The cicular segment:

It is a part of the circle's surface bounded by a chord and an arc. The chord divides the circle into a minor segment and a major segment (resembling a piece cut straight off the edge of a metal disc).





First: Area of the circular sector

The area of a circular sector can be calculated in three ways depending on the given data in the engineering drawing or design:

1. In terms of arc length and radius:

$A = \frac{1}{2} lr$

2. In terms of the centeral angle in radian:

Since, $L = \boldsymbol{\theta}^{\mathbf{rad}} \times \mathbf{r}$

Then , $A = \frac{1}{2} r^2 \, \theta^{\,rad}$

3. 2. In terms of the centeral angle in degrees:

Since, $\boldsymbol{\theta}^{\mathbf{rad}} = (\frac{\theta° \times 2\pi}{360°})$

Then , $A = \frac{\theta°}{360°} \times \pi r^2$

Example(1):

A circular sector has a radius of 16 cm and the measure of its central angle is 120°. Find its area to the nearest square centimeter.

Answer:
Since,

$$\theta° = 120° \text{ ، } r = 16$$

and $A = \frac{\theta°}{360°} \times \pi r^2$
then ,

$$\frac{120°}{360°} \times \pi 16^2 \approx 268\ cm^2$$

Example(2):

A surveillance camera lens on a production line scans a circular sector with a radius of 10 meters with a central angle of 1.5 radians. Calculate the area of the coverage zone.

Answer:

Since: r = 10 ، $\boldsymbol{\theta}^{\mathbf{rad}} = 1.5$

and A $= \frac{1}{2} r^2 \ \boldsymbol{\theta}^{\mathbf{rad}}$

then, A= $\frac{1}{2} \times 10^2 \times 1.5$

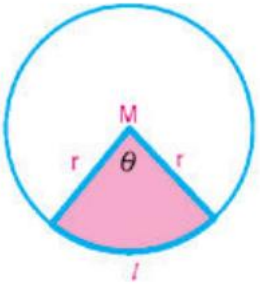
Example(3):

Find the area of the circular sector whose perimeter is 28 cm with a radius length 9 cm.

Answer:

Since: r = 9 ، perimeter $= 28$

Perimeter of the sector = $2r + l$



$$2r + l = 28$$

$$l = 28 - 18 = 10$$

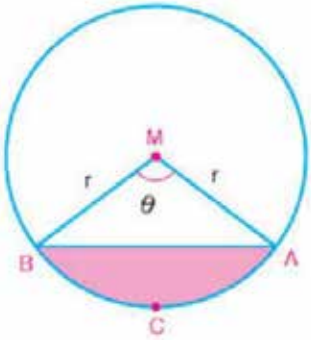
Area of the sector = $\frac{1}{2} lr =$

$$= \frac{1}{2} \times 10 \times 9 = 45 \text{ cm}^2$$

Second: Area of the circular segment

The area of a circular segment is calculated using a single formula that combines circular measure (radians) and the sine of the angle

A= $\frac{1}{2} r^2 \ (\boldsymbol{\theta}^{\mathbf{rad}} - \sin \boldsymbol{\theta})$



Example(1):

A metal rotating shaft has a radius of 10 cm. A portion of it (a circular segment) was cut using a milling machine to install a metal key. If the central angle corresponding to the cut is 90 degrees (which is approximately equivalent to 1.57radians, what is the area of the removed section?

Answer:

Since, r = 10 ، $\theta = 90°$ ، $\theta^{rad} = 1.57$

Then A= $\frac{1}{2}\, r^2\, (\boldsymbol{\theta^{rad}}$ **- sin** $\boldsymbol{\theta}$)

$= \frac{1}{2} \times 10^2 \times (1.57 - sin90)$

Since $\sin 90° = 1$

Then, A= $50 \times (1.57 - 1) = 50 \times 0.57 = 28.5$ cm^2

Example(2):

Find the area of the circuler segment whose circle radius is 8 cm and the measure of its angle is 150°

Answer:

r = 8 ،$\theta = 150°$ ،$\theta_{rad} = \frac{150 \times \pi}{180} = \frac{5\pi}{6}$

Then A= $\frac{1}{2}\, r^2(\theta_{rad} - sin\theta)$

$= \frac{1}{2} \times 8^2 \times \left(\frac{5\pi}{6} - sin150\right) =$ 67.7758 cm2 approximatly

Try to solve

1. A windshield wiper for a heavy-duty truck is 60 cm long and rotates to clean the glass through an angle of 120 degrees. Required: Calculate the area of the glass cleaned by the wiper in each full sweep. (Consider the wiper length as the radius, and use the formula for the area of a circular sector).

2. In a gearbox design, a gear with a radius of 12 cm rotates to drive another gear. If the first gear moves such that one of its teeth traces an arc path with a length of 18 cm:

-Calculate the gear's angle of rotation in radians.
-Calculate the area of the circular sector swept by the gear during this rotation.

3. In a gearbox design, a gear with a radius of 12 cm rotates to drive another gear. If the first gear moves such that one of its teeth traces an arc path with a length of 18 cm:

Calculate the gear's angle of rotation in radians.

Calculate the area of the circular sector swept by the gear during this rotation.

Check your understanding

Choose:

1. The area of a circular sector with a radius of 4 cm and an arc length of 6 cm equals….. cm^2

a) 10 b) 12 c) 24 d) 48

2. If the central angle of a circular segment in a wooden board with a radius of .50 cm is 60 degrees, find the area of this circular segment

3. True or False with Justification:

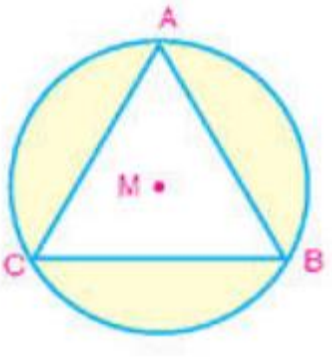
"A circular segment is always larger in area than the circular sector sharing the same arc."

Exercises

1.

A laser cutting machine is programmed to cut an acrylic sheet. The machine cut a circular sector with an area of $150\ cm^2$ and a central angle of 1.2 radians. Find the radius of the laser path.

2. In the opposite figure:



An equilateral triangle ABC is inscribed in a circle Mwith a radius of 8cm. Find the area of each of the shaded circular segments.

3. Find the area of the major circular segment whose chord length equals the radius of its circle, which is 12cm.

4. Open Discussion Question: In the design of aircraft windows and pressurized metallic structures, why do engineers prefer cutting panels with "circular sectors and segments" (rounded curves) instead of sharp angles? How does today's lesson help us calculate the exact amount of metal required for these curved components?

The unit Project

In your specialization, use what you have learned about complex numbers and equations to prepare a project according to the following specifications.

Project Assessment Form

School: .....

Project Title: .....

Date: .... / .... / .....

Project Participants: .....

Project Rubric

M	Criterion	Beginning	Intermediate	Advanced	Proficient	Score
1	Problem Identification and Overall Project Treatment	Attempts are made, but the problem does not appear to be clear.	The minimum required for understanding the problem appears, without sufficient evidence.	Demonstrates understanding of the topic, but the evidence is not clear or sufficient.	Demonstrates a correct and clear understanding of the problem and presents multiple pieces of evidence.	
2	Availability of Creative Solutions to the Problem	Is unable to generate strategies to solve the problem.	Begins but fails to complete the solution.	Begins in a reasonable way but leaves some parts of the task incomplete.	Uses a clear and organized approach and demonstrates high-level creative skills.	
3	Students' Presentation Skills	Needs guidance in organizing the presentation. The points are presented randomly, making them difficult to follow.	Organization is weak and the sequence does not facilitate understanding.	Organization or logical sequencing is present, but the presentation does not always follow a clear structure.	Includes a clear introduction, main topic, and conclusion. The points are presented in a logical sequence using appropriate language.	

M	Criterion	Beginning	Intermediate	Advanced	Proficient	Score
	Final Score					

Committee Decision:

.....

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Signature: .....

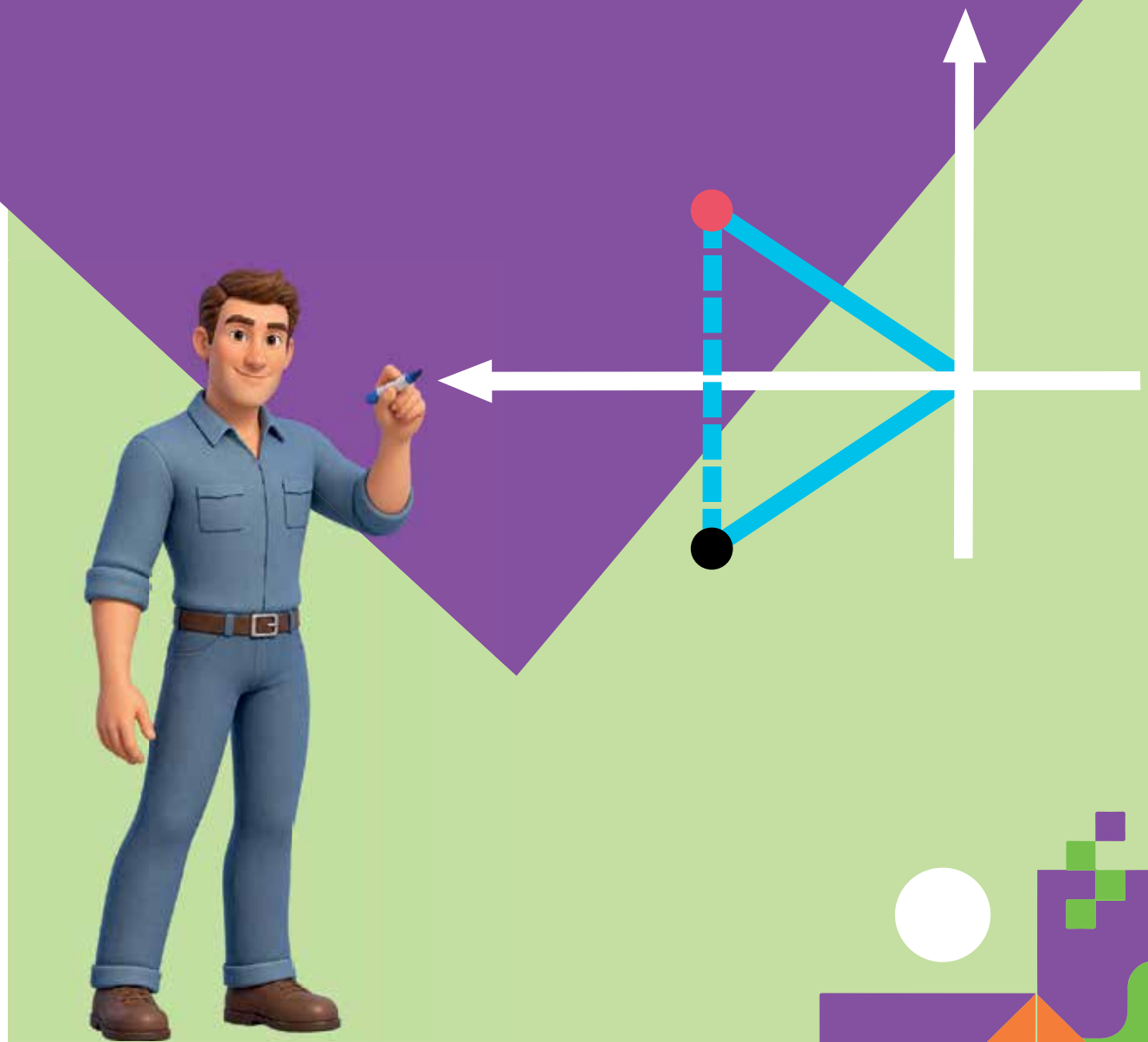
Unit Three
Complex Numbers and Quadratic Equations

Lesson 1: Complex Numbers

Lesson 2: Addition and Subtraction of Complex Numbers

Lesson 3: Multiplication the Complex Numbers – Conjugate Numbers

Lesson 4: Determining the Types of Roots of a Quadratic Equation

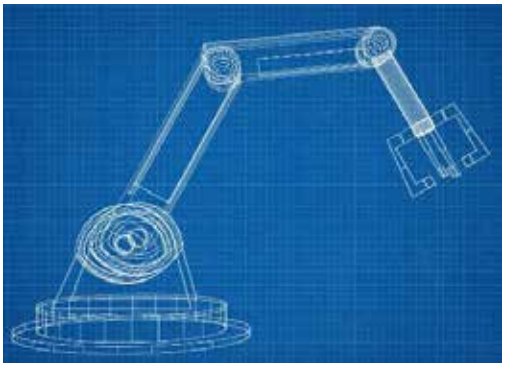


Lesson One: Complex Numbers

Learning Outcomes

- Recognize the concept of the imaginary number, its different powers, and simplify them.
- Identify the real and imaginary parts of a complex number accurately.
- Derive the conditions for equality of two complex numbers and apply them in solving equations.

Think and Discuss!



If you were working on a smart production line and faced a mechanical control system that required solving the equation:

$$x^2 + 16 = 0$$

Can this problem be solved in the set of real numbers? If imaginary roots appear, what does that mean for the stability of the robot's motion?

Learn

Concept of the Imaginary Number

An imaginary number is a number whose square is equal to (−1), and it is denoted by the symbol i.

By definition: $i^2 = -1$, and therefore $i = \sqrt{-1}$

Powers of the Basic Imaginary Number

Fourth Power	Third Power	Second Power	First Power
$i^4 = 1$	$i^3 = -i$	$i^2 = -1$	$i^1 = i$

Example (1): Simplifying Powers

Simplify: i^{22}

Solution: Divide the exponent 22 by 4. The remainder is 2, so $i^{22} = i^2 = -1$

Example (2): Negative Radicand

Find the value of $\sqrt{-25}$ in the set of imaginary numbers.

Solution: $\sqrt{-25} = \sqrt{(-25 \times -1)} = \sqrt{25} \times \sqrt{-1} = 5i$

Example (3): Solving an Equation

Solve the equation $x^2 + 9 = 0$ in the set of imaginary numbers.

Solution: $x^2 = -9$, hence $x = \pm\sqrt{(-9)}$, so $x = \pm 3i$

Try to Solve (1)

Write each expression in its simplest form:

- (a) i^{36}
- (b) $\sqrt{-16}$
- (c) Solve $x^2 + 4 = 0$

Concept of the Complex Number

A complex number is a number that can be written in the form:

$$Z = a + bi$$

where a and b are real numbers and $i^2 = -1$.

- a is called the real part.
- b is called the imaginary part.

Applied Examples

Example (1): Identifying the Parts

For the complex number $Z = 4 - 3i$, identify the real part and the imaginary part.

Solution: Real part = 4; Imaginary part = −3

Example (2): Standard Form

Write $\sqrt{(-4)} + 5$ in the form $Z = a + bi$.

Solution: Since $\sqrt{(-4)} = 2i$, the complex form is $Z = 5 + 2i$.

Example (3): Application in Car Maintenance

If the test signal for the car's small control circuit is given by $Z = -2i + 7$, determine the real resistance component.

Solution:

Writing the number in standard form, the resistance component is the real part, so the real resistance is 7 Ω.

Try to Solve (2)

Identify the real and imaginary parts of each complex number:

- (a) $Z = -3 + 5i$
- (b) $Z = \sqrt{(-49)} + 2$

Equality of Two Complex Numbers

Two complex numbers are equal if and only if their real parts are equal and their imaginary parts are equal.

$$\text{If } a + bi = c + di, \text{ then } a = c \text{ and } b = d.$$

Applied Examples

Example (1): Finding Unknowns Directly

Find x and y if $x + yi = 5 - 2i$.

Solution: By comparing the real and imaginary parts directly: $x = 5$, $y = -2$.

Example (2): Solving Linear Equations

Find the real values of x and y that satisfy: $(2x - 3) + 4yi = 7 + 12i$.

Solution:

From the real parts: $2x - 3 = 7$, so $x = 5$.

From the imaginary parts: $4y = 12$, so $y = 3$.

Example (3): Control and Stability Systems

In a robotic-arm mechanical load-control system, the relation is $(x + y) + (x - y)i = 6 + 2i$. Find the real constants x and y.

Solution:

Equating the parts gives:

1) $x + y = 6$
2) $x - y = 2$.

Adding the two equations: $2x = 8$, so $x = 4$.

Substituting in $x + y = 6$ gives $y = 2$.

Check Your Understanding:

Find the real values of x and y if $(x + 1) + 2yi = 5 - 8i$

Exercises on Complex Numbers

Question One: Choose the Correct Answer

1. The value of i^{44} is:

(A) i (B) −i (C) 1 (D) −1

2. The imaginary part of Z = 5 − 2i is:

(A) 5 (B) 2 (C) −2 (D) −2i

Question Two: Industrial Applications

3. In the control system of a smart industrial robot inside a production line, if the feedback signal is modeled by (A − 7) + (B + 3)i = 0, find the constant values A and B required to prevent excessive vibration and maintain the robot-arm stability.

4. Find the total electric current in a parallel circuit if the current in the first branch is (4 − 2i) A and the current in the second branch is $\frac{6+3i}{2+i}$ A.

Lesson Two: Addition and Subtraction of Complex Numbers

Learning Outcomes

- Identify the real and imaginary parts of a complex number.
- Derive the rules for adding and subtracting complex numbers algebraically.
- Addition and subtraction of complex numbers are performed efficiently.
- Apply addition and subtraction of complex numbers in suitable technological applications.

Think and Discuss

Solar Power Generation Stations



The importance of adding and subtracting complex numbers becomes clear in understanding electronic circuits, wireless communication systems, and control systems in factories.

Suppose two suitable units are connected to a solar power generation station. If the first unit produces a current of $3 + 2i$ A and the second unit produces a current of $5 - 3i$ A, how can you find the total current resulting from combining the two units?

Learn

Rule for Adding Complex Numbers

When adding two complex numbers, add the real parts together and the imaginary parts together.

The commutative and associative properties can be used when adding complex numbers.

Example (1)

Find the sum of the two complex numbers: $5 + 3i$ and $4 + 2i$.

Solution:

$(5 + 3i) + (4 + 2i) = (5 + 4) + (3 + 2)\, i = 9 + 5i.$

Example (2): Agricultural Application



In one of the smart agricultural greenhouses, there are two pumps connected in parallel. If the electrical current supplied to the first pump can be represented by $4 + i$ A, while the current supplied to the second pump is represented by $3 - 2i$ A, find the total current supplied to the irrigation network.

Solution:

$(4 + i) + (3 - 2i) = (4 + 3) + (1 - 2)i = 7 - i$

Rule for Subtracting Complex Numbers

When subtracting two complex numbers, subtract the real parts and the imaginary parts separately.

Apply the sign rule when removing parentheses.

Example (3)

Find the result of subtracting the complex number $5 + 3i$ from $6 + 7i$.

Solution:

$(6 + 7i) - (5 + 3i) = 6 + 7i - 5 - 3i = (6 - 5) + (7 - 3)i = 1 + 4i.$

Example (4): Electrical Application



In one of the alternating-current circuits connected in series, the electrical impedance of the first coil is $8 + 6i\ \Omega$, and the impedance of the second coil is $2 - 4i\ \Omega$. Find:

1) The total impedance of the circuit.

2) The difference between the impedance of the first coil and the impedance of the second coil.

Solution:

1) $(8 + 6i) + (2 - 4i) = (8 + 2) + (6 - 4)i = 10 + 2i$.

2) $(8 + 6i) - (2 - 4i) = 8 + 6i - 2 + 4i = (8 - 2) + (6 + 4)i = 6 + 10i$.

Try to Solve

Find the result of each expression in its simplest form:

- a. $(5 + 3i) + (2 + 4i)$
- b. $(7 - 2i) - (4 - 6i)$

Critical Thinking



When checking a wireless communication radar system, the transmitted signal was represented by the complex number $8 - 5i$ microvolts. When the reflected signal returned, it had decreased by $3 - 2i$ microvolts. What is the net signal value recorded by the device?

Check Your Understanding

If you have the two complex numbers: $6 - 2i$ and $-3 + 5i$, find their sum and their difference, respectively.

Exercises on Addition and Subtraction of Complex Numbers

1. Complete the following:

- a. In the complex number $9 - 11i$, the real part is ____ and the imaginary part is ____.
- b. If $(2 + 3i) - (a + bi) = 5 + 4i$, then a = _____, b = _____.
- c. What is the value of $(i - 1) + (i + 1)$?

2. Choose the correct answer:

a. What is the result of $(5i - 3) + (2i + 10)$?

(A) $3i - 7$ (B) $13 - 7i$ (C) $3i + 7$ (D) $13 + 7i$

b. If the total impedance of an electrical circuit is $9 + 5i$ and the impedance of the first part is $4 + i$, then the impedance of the second part is:

(A) $13 + 6i$ (B) $5 + 4i$ (C) $4i - 5$ (D) $-4i - 5$



3. **Device Maintenance:** If you have two measuring devices connected in series, where the voltage of the first device is $12 + 5i$ volts and the voltage of the second device is $8 - 9i$ volts. Find the total voltage for the two devices.

4. **Hotel Events:** In a conference hall at a hotel, a speaker emitted a sound wave represented by the complex number $6 + 12i$ decibels. To prevent echo, an engineer operates a smart system to send an opposite wave of $8 + 4i$ decibels. Find the net sound signal remaining in the hall.



Lesson Three: Multiplication of Complex Numbers – Conjugate Numbers

Learning Outcomes

- Relate operations on complex numbers to their technological applications.
- Perform multiplication of complex numbers accurately.
- Recognize the concept of conjugate numbers and their properties.
- Use the property of multiplying conjugate numbers to simplify expressions and division.
- Apply problem-solving skills in realistic industrial and technological situations.

Think and Discuss



Technological Situation: In the control panel of an industrial robot, we have two impedances:

$$Z_1 = 2 + 3i \quad \text{and} \quad Z_2 = 4 - i$$

- 1. How can you multiply these two quantities together?
- 2. What happens when we multiply a complex number by its conjugate?
- 3. What is the engineering or practical benefit of this property in electrical circuits?

Enrichment Information: Impedance, denoted by the symbol Z

The total impedance that appears in a circuit when ordinary resistance (R) is combined with other elements such as inductive coils (L) and capacitors (C) is called impedance and is represented by Z. Because the response of coils and capacitors is affected by frequency, their values are represented using a complex number containing a real part and an imaginary part: Z = R + Xi.

Multiplication of Complex Numbers

If we have two complex numbers in standard form:

$$Z_1 = a + bi \quad , \quad Z_2 = c + di$$

Then their product is: $Z_1 \times Z_2 = (a + bi)(c + di)$

$$= ac + adi + bci + bdi^2$$

Since $i^2 = -1$, we obtain:

$$Z_1 \times Z_2 = (ac - bd) + (ad + bc)i$$

Steps for Multiplication

1. Write the two numbers in standard form (a + bi).
2. Apply the distributive property (first × first, first × second, …).
3. Replace i^2 by −1.
4. Add the real terms together and the imaginary terms together.

Important Note

- $i^2 = -1$
- $i^3 = i^2 \cdot i = -1 \cdot i = -i$
- $i^4 = (i^2)^2 = (-1)^2 = 1$

Example (1)

Find the product of the two complex numbers (3 + 2i)(1 + 4i).

Solution:

$(3 + 2i)(1 + 4i) = 3 \times 1 + 3 \times 4i + 2i \times 1 + 2i \times 4i$

$= 3 + 12i + 2i + 8i^2$

$= 3 + 14i + 8(-1) = 3 + 14i - 8 = -5 + 14i.$

Example (2): Industrial Application

The voltage in an alternating-current circuit is V = 5 − 2i volts, and the current is I = 2 + 3i amperes. Calculate the complex power P = V × I.

Solution:

$$P = (5 - 2i)(2 + 3i)$$

$$= 5 \times 2 + 5 \times 3i + (-2i) \times 2 + (-2i) \times 3i$$

$$= 10 + 15i - 4i - 6i^2$$

$$= 10 + 11i - 6(-1)$$

$$= 10 + 11i + 6$$

$$= 16 + 11i \text{ (volt-ampere)}$$

Conjugate Numbers

Definition of the Conjugate of a Complex Number

The conjugate of the complex number $Z = a + bi$ is the complex number that has the same real part and the opposite sign of the imaginary part.

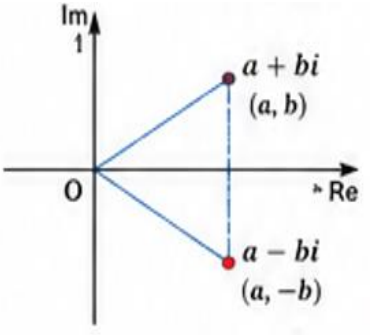
$$\text{If } Z = a + bi, \text{ then its conjugate is: } \bar{Z} = a - bi.$$

Examples

- The conjugate of $5 + 3i$ is $5 - 3i$.
- The conjugate of $7 - 2i$ is $7 + 2i$.
- The conjugate of $-4i$ is $4i$.
- The conjugate of the real number 8 is 8.

Geometric Representation

On the Argand plane (complex plane), a complex number and its conjugate are reflections of each other across the real axis (the x-axis).



- The real part (a) remains constant.
- The imaginary part (b) changes its sign.

Property of Multiplying Conjugate Numbers

When a complex number is multiplied by its conjugate, we always obtain a positive real number.

$$Z \times \bar{Z} = (a + bi)(a - bi) = a^2 - b^2i^2 = a^2 + b^2$$

Therefore:

$$(a + bi)(a - bi) = a^2 + b^2 \text{ (a positive real number)}$$

Examples

- $(4 + 3i)(4 - 3i) = 4^2 + 3^2 = 16 + 9 = 25$

- $(1 - i)(1 + i) = 1^2 + 1^2 = 2$
- $(2 + 5i)(2 - 5i) = 2^2 + 5^2 = 4 + 25 = 29$

Important Benefit

This property helps us divide complex numbers and eliminate the imaginary number from the denominator, making the denominator a real number.

Industrial and Technological Applications

1. Removing the Imaginary Number from the Denominator

In parallel circuits, we need to calculate total impedance. The result is often in fractional form, so we multiply the numerator and denominator by the conjugate.



Applied Example: CNC Machine Control Panel

Suppose the impedance is $Z = \frac{10}{3+i}$. Write it in standard form.

Solution:

$$Z = \frac{10}{3+i} \times \frac{3-i}{3-i} = \frac{10(3-i)}{(3+i)(3-i)}$$

$$= \frac{30-10i}{3^2+1^2} = \frac{30-10i}{10} = 3 - i$$

Reminder

Real resistance: R = 3 Ω. Reactance: X = −1 Ω.

2. Calculating Power in Alternating-Current Circuits

Complex power is given by S = V × I*, where I* is the conjugate of the current.



It is used to

calculate active power, reactive power, and the power factor.

Try to Solve

Exercise 1: Find the product in its simplest form:

- $(2 - 3i)(4 + i)$
- $(1 + 5i)^2$

Exercise 2: In a robot arm control system, $Z = (3 - 2i)(3 + 2i)$. Find the value of Z and state whether the resulting number is real or imaginary.

Exercise 3: Write the following in standard form using the conjugate:

$$\frac{5 + 5i}{2 - i}$$

Check Your Understanding

Working with your classmates in the electricity or electronics workshop, write a one-paragraph report explaining how the property of multiplying conjugate numbers helps simplify calculations in electrical circuits and convert imaginary denominators into real quantities that can be easily measured with an ohmmeter.

Exercises on Multiplication of Complex Numbers – Conjugate Numbers

First: Choose the Correct Answer

1. The product of the complex number (2 − i) and its conjugate is:

(A) −3 (B) 5 (C) 3 (D) 1 + 4i

2. If Z = 3i, then Z × $\overline{Z}$ equals:

(A) −9 (B) 9 (C) 0 (D) 3

3. The expression (2 + i)(2 − i) represents a geometric application that gives a real value equal to:

(A) 5 (B) 3 (C) 4 – i (D) 0

Second: Find the Product

- (8 + 2i)(9 − i)
- (9 + 2i)(10 − i)
- (10 + 2i)(11 − i)

Production and Industry Problems

1. In hybrid automotive engineering, if two impedance signals are $S_1 = -2 + 4i$ and $S_2 = 1 - 3i$, find the total complex magnitude resulting from multiplying them.

2. In a hydraulic control system, an interference signal has the value $\frac{13}{2+3i}$ Simplify the interference by converting the fraction to standard form.

Lesson Four: Determining the Types of Roots of a Quadratic Equation

Learning Outcomes

- Determine the type of the roots of a quadratic equation using the discriminant (Δ).
- Relate the mathematical concept to applications such as robotics (arm motion), motors (rotation angles), and circuits (calculating speed ratios).
- Apply solution steps in realistic industrial situations.

Think and Discuss



Imagine that a sprinkler used to irrigate agricultural land operates according to a quadratic equation that determines the irrigation range. If we want to know whether the water reaches a specific point, we need to determine the type of the equation's roots: Are they real? Are they equal? Or are they non-real? What does each case mean?

Learn

Standard Form of a Quadratic Equation

$$ax^2+bx+c=0, \qquad (a\neq 0,\ a,b,c \in R)$$

The Discriminant (Δ)

$$\Delta=b^2-4ac$$

Determining the Type of Roots

- If $\Delta > 0$, there are two distinct real roots.
- If $\Delta = 0$, there are two equal real roots.
- If $\Delta < 0$, there are no real roots (the roots are complex/non-real).

Steps for Determining the Types of Roots

1. Identify the coefficients a, b, and c.
2. Calculate the discriminant Δ.
3. Determine the type of roots according to the value of Δ.

Example (1): Determine the Type of Roots for Each Equation

1. $x^2 - 5x + 6 = 0$

Solution: $a = 1, b = -5, c = 6$.

$\Delta = (-5)^2 - 4(1)(6) = 25 - 24 = 1$, so $\Delta > 0$.

Therefore, there are two distinct real roots.

2. $x^2 - 4x + 4 = 0$

Solution: $a = 1, b = -4, c = 4$.

$\Delta = (-4)^2 - 4(1)(4) = 16 - 16 = 0$.

Therefore, there are two equal real roots.

3. $x^2 + x + 1 = 0$ (and find its roots)

Solution: $a = 1, b = 1, c = 1$.

$\Delta = (1)^2 - 4(1)(1) = 1 - 4 = -3$, so $\Delta < 0$.

Therefore, there are two complex (non-real) roots.

To find the roots, substitute in the quadratic formula: $\frac{-b \pm \sqrt{\Delta}}{2a}$

$$\frac{-1 \pm \sqrt{3i^2}}{2 \times 1} = \frac{-1 \pm \sqrt{3}i}{2}$$

Summary Table

Example	Value of Δ	Type of Roots
$x^2 - 5x + 6 = 0$	$\Delta > 0$	Two distinct real roots
$x^2 - 4x + 4 = 0$	$\Delta = 0$	Two equal roots
$x^2 + x + 1 = 0$	$\Delta < 0$	Two non-real complex roots

Try to Solve

1. Determine the type of roots of the equation: $x^2 + 3x + 2 = 0$.

2. Robot-arm motion equation: $x^2 - 6x + 9 = 0$. What is the type of roots?

3. Determine the type of roots of the equation: $6x^2 = 19x - 15$.

4. Determine the type of roots of the equation: $x(x + 3) = -5$.

5. Prove that the roots of the equation $-x^2 + 5x - 30 = 0$ are complex and non-real, then find its roots.

Check Your Understanding

Choose the correct answer:

1. If $\Delta < 0$, then the roots are:

- (A) Equal real roots
- (B) Distinct real roots
- (C) Non-real roots

• Convert the equation $x^2 - 2x + 1 = 0$ to discriminant form and determine the type of roots.

Exercises on Determining the Types of Roots of a Quadratic Equation

1. A quadratic equation describing the motion of a car is $t^2 - 4t + 3 = 0$. Determine the type of roots.

2. Find the value of k that makes the roots of the equation $x^2 + 4x + k = 0$ two distinct real roots.

3. Determine the type of roots of the equation $x^2 - 14x + 49 = 0$.

4. Robot-arm motion equation: $x^2 - 6x + 5 = 0$. What is the type of roots?

5. Determine the type of roots of the equation $x^2 = 19x - 84$.

6. Determine the type of roots of the equation $x(x - 5) = 8$.

7. Prove that the roots of the equation $-x^2 + x - 5 = 0$ are complex and non-real, and find its roots.

Unit Project

In your specialization, use what you have learned about complex numbers and equations to prepare a project according to the following specifications.

Project Assessment Form

School: .....

Project Title: .....

Date: .... / .... / .....

Project Participants: .....

.....

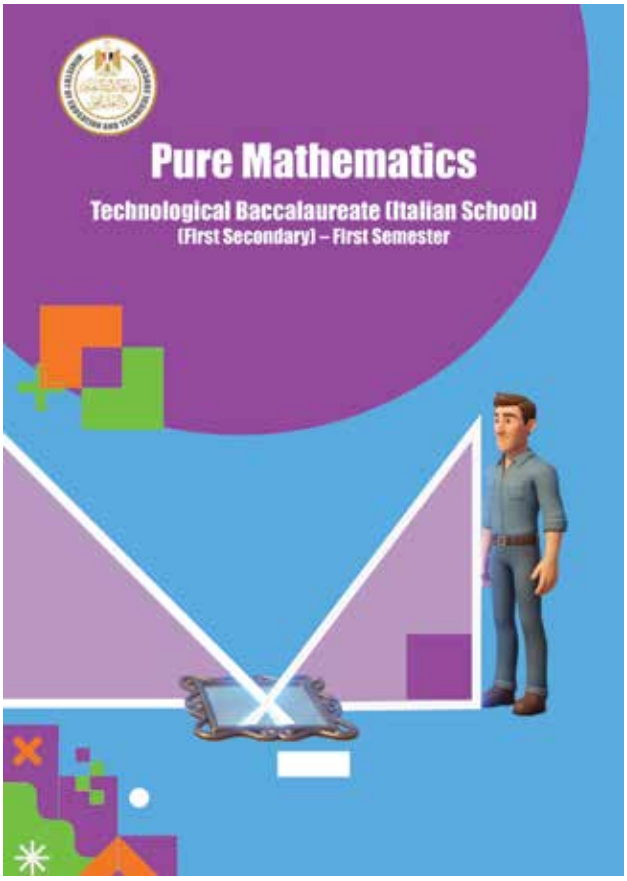
Project Rubric

M	Criterion	Beginning	Intermediate	Advanced	Proficient	Score
1	**Problem Identification and Overall Project Treatment**	Attempts are made, but the problem does not appear to be clear.	The minimum required for understanding the problem appears, without sufficient evidence.	Demonstrates understanding of the topic, but the evidence is not clear or sufficient.	Demonstrates a correct and clear understanding of the problem and presents multiple pieces of evidence.	
2	**Availability of Creative Solutions to the Problem**	Is unable to generate strategies to solve the problem.	Begins but fails to complete the solution.	Begins in a reasonable way but leaves some parts of the task incomplete.	Uses a clear and organized approach and demonstrates high-level creative skills.	
3	**Students' Presentation Skills**	Needs guidance in organizing the presentation. The points are presented randomly, making them difficult to follow.	Organization is weak and the sequence does not facilitate understanding.	Organization or logical sequencing is present, but the presentation does not always follow a clear structure.	Includes a clear introduction, main topic, and conclusion. The points are presented in a logical sequence using appropriate language.	
	Final Score					

Committee Decision:

.....

Signature: .....



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